Alexander Galozy

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Research Profile

Machine learning researcher specialising in sequential decision-making under uncertainty and partial observability. My work focuses on reinforcement learning and bandit models with latent state, studying how information structure and optimisation dynamics shape adaptive behaviour and robustness under distribution shift. I develop statistical and algorithmic frameworks for learning in stochastic dynamical systems, with with a growing focus on how information-conditioned behaviour is structured and what happens to it under standard learning operations.

Academic Appointments

2024–2026	**Postdoctoral Researcher**, Halmstad University, Sweden CAISR Health. PI: Jens Lundström Pursuing theoretical research in RL under partial observability. Contributing to interdisciplinary healthcare AI projects with Region Halland. Teaching and supervision in AI and data science courses.
2018–2023	**Doctoral Researcher (PhD in Artificial Intelligence)**, Halmstad University Advisor: Sławomir Nowaczyk Research on sequential decision models in partially observed dynamical systems. Participated in Vinnova-funded IMedA project on adaptive health interventions. Teaching and supplemental instruction in AI courses.

Education

Dec. 2023	**Ph.D. Artificial Intelligence**, Halmstad University, Sweden Thesis: *Mobile Health Interventions through Reinforcement Learning*
June 2018	**M.Sc. Artificial Intelligence**, Halmstad University, Sweden
May 2014	**Dipl.-Ing (FH), Mechatronics, Robotics and Automation** Hochschule für angewandte Wissenschaften Würzburg-Schweinfurt, Germany

Core Research Areas

RL & Bandits	Latent-state bandits, partial observability, adaptive decision processes, epistemic structure, information-seeking behaviour.
Learning Under Uncertainty	Distribution shift, robustness, behavioural equivalence, stochastic decision systems.
Optimisation & Decision Theory	Policy learning, exploration–exploitation trade-offs, learning dynamics, regret analysis.

Publications

Preprints

- Galozy, A. (2026). On the Structural Non-Preservation of Epistemic Behaviour under Policy Transformation. *arXiv:2602.21424*. Under review at RLC 2026.
- Galozy, A., Alawadi, S., Kebande, V., & Nowaczyk, S. (2023). Beyond Random Noise: Insights

on Anonymization Strategies from a Latent Bandit Study. *arXiv:2310.00221*.

Peer-Reviewed Journal Articles

- Galozy, A. & Nowaczyk, S. (2023). Information-gathering in latent bandits. *Knowledge-Based Systems*.
- Galozy, A., Nowaczyk, S., & Ohlsson, M. (2025). A new bandit setting balancing information from state evolution and corrupted context. *Data Mining and Knowledge Discovery*.
- Galozy, A. & Nowaczyk, S. (2020). Prediction and pattern analysis of medication refill adherence through electronic health records. *Journal of Biomedical Informatics*.
- Galozy, A., et al. (2020). Pitfalls of medication adherence approximation through EHR and pharmacy records. *International Journal of Medical Informatics*.
- Etminani, K., et al. (2021). Improving medication adherence through adaptive digital interventions. *JMIR Research Protocols*.
- Jendle, J., et al. (2023). Patterns and predictors associated with long-term glycemic control. *Journal of Diabetes Science and Technology*.
- Agvall, B., et al. (2024). Factors influencing hospitalization or emergency visits in type 2 diabetes. *Cardiovascular Diabetology*.

Conference Papers

- Galozy, A., Nowaczyk, S., & Sant'Anna, A. (2019). Towards understanding ICU treatments using patient health trajectories. *KR4HC*.

Publication metrics: 10+ peer-reviewed publications; 8 as first author; 116 citations; h-index: 5.

Technical Expertise

Core Areas	Reinforcement learning, bandits, sequential decision-making, probabilistic modelling, learning under uncertainty.
Methods	Latent-state modelling, stochastic processes, time-series modelling, deep learning, regret analysis.
Programming	Python, PyTorch, TensorFlow, Scikit-learn, Stable Baselines, XGBoost.
Computing	GPU/CUDA, large-scale training workflows, reproducible ML pipelines, experiment tracking.

Teaching & Supervision

2023–2025	**Course Responsible & Co-Developer** — Data Science: Theory, Practice and Societal Implications, Halmstad University.
2025–2026	**Lecturer** — Artificial Intelligence / AI for Health.
2025–2026	**Teaching Assistant** — Programming / Smart Health with Applications.
2017–2021	**Supplemental Instruction Leader** — Artificial Intelligence.
	Supervised $\geq$5 MSc and BSc theses in machine learning and applied AI.

Research Collaboration & Funding

IMedA	Vinnova-funded project on adaptive digital health interventions. Led methodological development using probabilistic and sequential models on large-scale longitudinal EHR data (Region Halland).
REMIND	EU-funded academia–industry exchange on healthcare AI and medical device validation.
AI.m	AI consultant for regional industry partners (Visible Care, Remainly, Area Academy).

ACADEMIC SERVICE

TPC Member	FMEC (Fog and Mobile Edge Computing), 2024–2026.
Reviewer	ECML-PKDD 2025–2026; International Journal of Data Science and Analytics.